



NORTH SYDNEY BOYS HIGH SCHOOL

2019 HSC ASSESSMENT TASK 3 (TRIAL HSC)

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using blue or black pen
- Board approved calculators may be used
- All necessary working should be shown in every question
- Each new question is to be started on a **new page**.
- Attempt all questions

Class Teacher:

(Please tick or highlight)

- ☐ Mr Berry
- ☐ Mr Ireland
- ☐ Dr Jomaa
- ☐ Ms Lee
- ☐ Mr Lin
- ☐ Ms Ziazaris

Student Number:

(To be used by the exam markers only.)

Question No	1-10	11	12	13	14	Total	Total
Mark	$\overline{10}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{15}$	$\overline{70}$	$\overline{100}$

Section I

10 marks

Attempt Questions 1-10

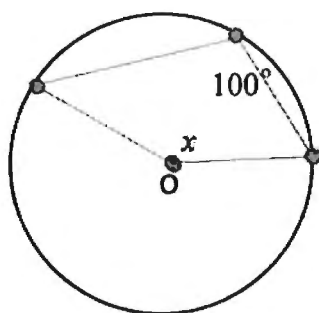
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1-10

1. The acute angle between the lines with equations $y = 3x - 2$ and $2x - 3y + 1 = 0$ is

- A. $39^{\circ}17'$
- B. $37^{\circ}52'$
- C. $105^{\circ}15'$
- D. $74^{\circ}45'$

2. Find the value of x



- A. 80°
- B. 160°
- C. 100°
- D. 200°

3. $\int \sin x \cos x dx =$

A. $\frac{1}{2} \sin 2x + C$

B. $-\cos x \sin x + C$

C. $\frac{1}{2} \sin^2 x + C$

D. $\frac{1}{2} \cos^2 x + C$

4. Differentiate $x^3 \log(\sqrt{x})$

A. $\frac{3x^2 \log x}{2} + \frac{x^2}{2}$

B. $\frac{1}{2} x^2$

C. $\frac{3x^2 \log x - x^2}{(\log x)^2}$

D. $\frac{5}{2} x^{\frac{3}{2}}$

5. A particle moving in simple harmonic motion has equation $x = 2 \sin \frac{1}{2} t$ where x is the displacement of the particle after t seconds.

Find the speed of the particle when $x = \sqrt{3}$.

A. $\frac{1}{4}$

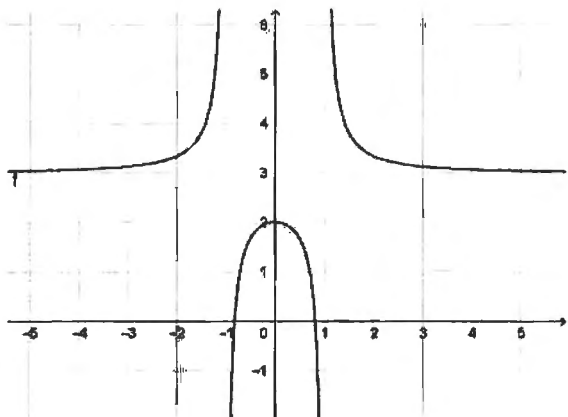
B. 1

C. $\frac{1}{2}$

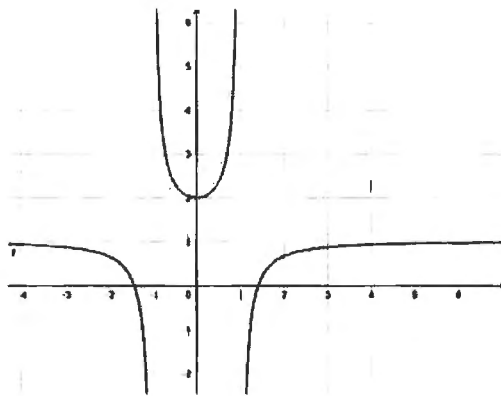
D. 2

6. The graph of $y = 2 - \frac{x^2}{1-x^2}$

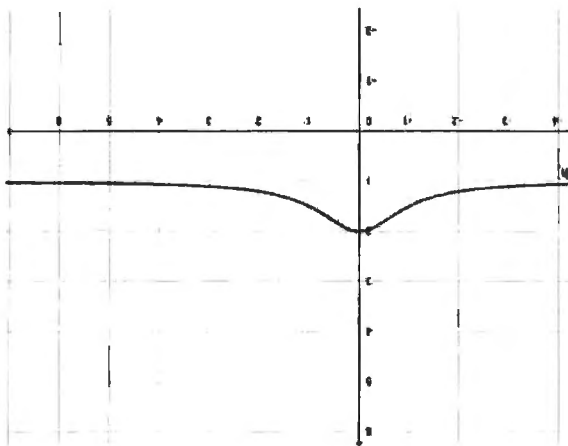
A.



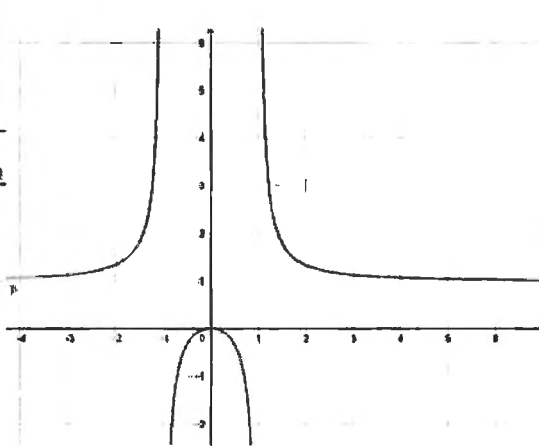
B.



C.



D.



7. The general solution of $\sin 2\theta = \sin \theta$ is

A. $2n\pi \pm \frac{\pi}{3}$

B. $2n\pi \pm \frac{\pi}{6}$

C. $n\pi$ or $2n\pi \pm \frac{\pi}{3}$

D. $n\pi$ or $2n\pi \pm \frac{\pi}{6}$

8. Evaluate $\lim_{x \rightarrow 0} \frac{\sin \frac{x}{2}}{3x}$

A. 6

B. $\frac{1}{3}$

C. $\frac{2}{3}$

D. $\frac{1}{6}$

9. If $y = \log_{1/a}(\frac{1}{N})$, $a > 0, N > 0$ then which one of the following is true.

A. $y = \log_a(-N)$

B. $y = \log_a(N)$

C. $y = \log_{-a}(\frac{1}{N})$

D. $y = \frac{\log_{1/a} 1}{\log_{1/a} N}$

10. The polynomial $P(x) = x^3 + px^2 + qx + r$ has real roots $\sqrt{m}, -\sqrt{m}$ and n . Find the value of pq .

A. 0

B. r

C. $-mn$

D. $-r$

Section II

60 marks

Attempt questions 11-14

Allow about 1 hour and 45 minutes for this section

Answer each question in a SEPARATE writing booklet. Extra writing booklets are available.

In Questions 11-14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks) Use a SEPARATE writing booklet.

- (a) A, B are the points $(-4,6)$ and $(1,-7)$ respectively. Find the coordinates of the point P dividing AB externally in the ratio 2:3. 2
- (b) Solve $\frac{3x+4}{x-5} \geq 2$ 3
- (c) Find the domain and range of $y = 2 \sin^{-1} \frac{x}{3}$ 2
- (d) Differentiate $\sqrt{\sec x}$ 2
- (e) $\int_4^{16} \frac{1}{\sqrt{x}(4+x)} dx$ using $u = \sqrt{x}$. Leave answer in exact form. 3
- (f) $\int_0^{\frac{\pi}{2}} \cos^2 3x dx$ Leave answer in exact form. 3

Question 12 continues on the next page.....→

Question 12 (15 marks) Use a SEPARATE writing booklet.

(a) Find the values of a for which $(x - a)$ is a factor of $x^2 - (a + 2)x + 6$ 1

(b) (i) Express $7 \cos \theta - \sin \theta$ in the form $R \cos(\theta + \alpha)$

where $R > 0$ and $0^\circ \leq \alpha \leq 360^\circ$. 2

(ii) Hence, solve $7 \cos \theta - \sin \theta = 5$ for $0^\circ \leq \theta \leq 360^\circ$ 2

(c) If x_1 is an approximation to the value $\sqrt[4]{m}$, show that a closer

approximation is $\frac{3x_1^4 + m}{4x_1^3}$, using Newton's Method. 2

(d) The acceleration of a particle moving in a straight line is given by

$$\frac{d^2x}{dt^2} = 2x - 3$$

where x is the displacement, in metres, from the origin O and t is the time in seconds. Initially the particle is at rest at $x = 4$.

(i) If the velocity of the particle is v m/s, show that $v^2 = 2(x^2 - 3x - 4)$. 2

(ii) Show that the particle does not pass through the origin. 1

(iii) Determine the position of the particle when $v = 10$. Justify your answer. 2

(e) Use the principle of mathematical induction to prove if n is a positive integer, that

$n(n + 3)$ is divisible by 2 3

Question 13 continues on the next page.....➡

Question 13 (15 marks) Use a SEPARATE writing booklet.

(a) $\int \frac{dx}{\sqrt{9-4x^2}}$

1

- (b) In a certain University course, the rate of students dropping out of the course was proportional to the number N , still attending, in excess of 400. This rate can be expressed by the equation $\frac{dN}{dt} = k(N - 400)$, where t is the number of weeks and k is constant.

(i) Show that $N = 400 + Ae^{kt}$ is a solution of this differential equation.

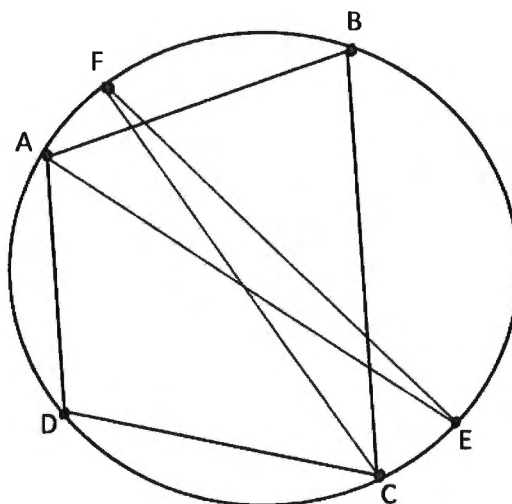
1

(ii) If 1000 students started the course and 10% had dropped out after 12 weeks, estimate the number of weeks it will take for half the students to remain.

2

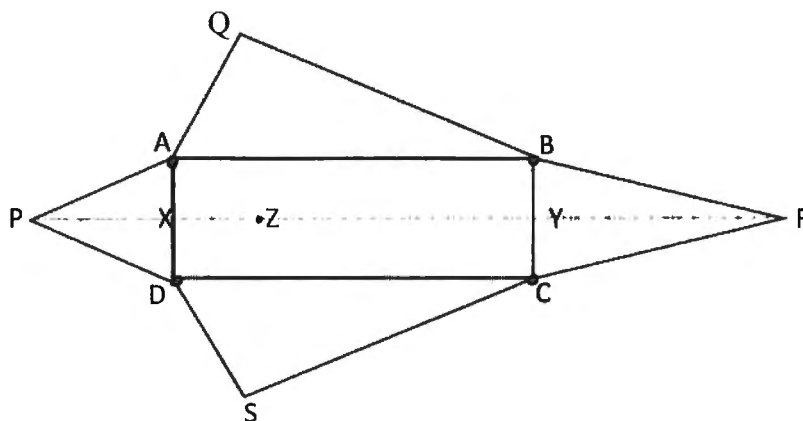
- (c) ABCD is a cyclic quadrilateral. The bisectors of the angles at A and C meet the circle again at E and F respectively.

3



Question 13 continues on the next page.....▶

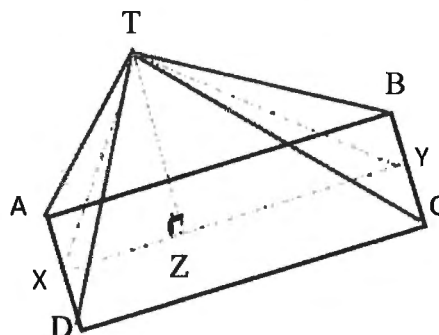
(d)



The figure shows the net of an oblique pyramid with a rectangular base. In this figure, $PXZYR$ is a straight line, $PX = 15\text{cm}$, $RY = 20\text{cm}$, $AB = 25\text{cm}$, and $BC = 10\text{cm}$.

Further, $AP = PD$ and $BR = RC$.

When the net is folded, points P, Q, R and S all meet at the apex T , which lies vertically above the point Z in the horizontal base, as shown below.



(i) Show that $\triangle TXY$ is right angled. 1

(ii) Hence show that T is 12cm above the base. 2

(iii) Hence find the angle that the face DCT makes with the base. 1

(e)(i) Sketch the parabola whose parametric equations are $x = t$ and $y = t^2$.

On your diagram mark the points P and Q which correspond to $t = -1$ and $t = 2$ respectively. 2

(ii) Show that the tangents to the parabola at P and Q intersect at $R(\frac{1}{2}, -2)$. 1

(iii) Let $T(t, t^2)$ be the point on the parabola between P and Q such that the tangent at T meets QR at the midpoint of QR . Let the midpoint be M . Show that the tangent at T is parallel to PQ . 1

Question 14 continues on the next page.....➡

Question 14 (15 marks) Use a SEPARATE writing booklet.

(a) A particle moves such that $x = \frac{3(\cos 2t - \sin 2t)}{\sqrt{2}}$ where x represents displacement in metres and t represents time in seconds.

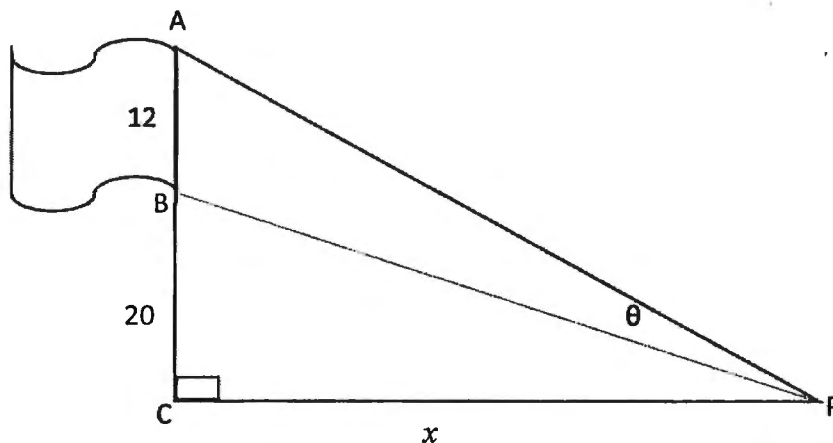
(i) Show that the particle is moving in simple harmonic motion. 2

(ii) Find the period of the motion. 1

(iii) Show that $x = 0$ when $t = \frac{\pi}{8}$. 1

(iv) Find the amplitude of the motion. 2

(b) A flagpole 12 m high stands on the top of a building 20 m high. The flagpole subtends an angle of θ at a point P on the ground and the distance from P to the building is x metres.

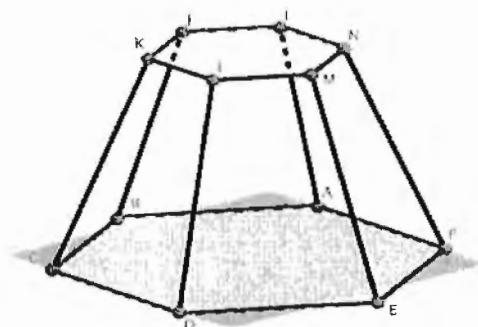


(i) Show that $\theta = \tan^{-1}\left(\frac{12x}{x^2 + 640}\right)$ 2

(ii) Find x that makes θ a maximum. 3

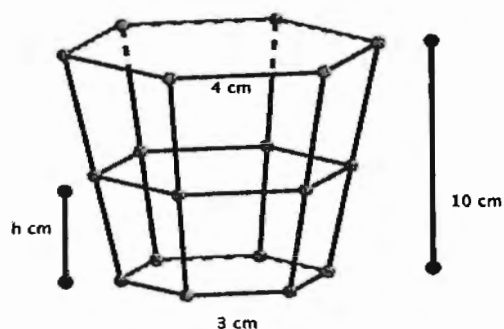
Question 14 continues on the next page.....→

- (c) A frustum of height H is made by cutting off a hexagonal pyramid of base IJKLMN with side length b from a regular hexagonal pyramid of base ABCDEF with side length a . It is given that the volume of the frustum is $\frac{\sqrt{3}}{2}(a^2 + ab + b^2)H$



An empty glass in the form of an inverted frustum described above with the height 10 cm, the side of the base and the top are 3 cm and 4 cm respectively. Water is being poured into the glass. Let h be the depth of the water inside the glass at time t seconds where

$$0 \leq h \leq 10.$$



- i. Show that the volume $V \text{ cm}^3$ of water inside the glass at time t is given by 2

$$V = \frac{\sqrt{3}}{2000} (27000h + 900h^2 + 10h^3)$$

- ii. If the volume of the water in the glass is increasing at the rate $7 \text{ cm}^3/\text{s}$, find the rate of increase of depth of water at the instant when $h=5\text{cm}$. 2

END OF EXAMINATION

Yr12 Solutions Ext1 Task 3 2019

M/C

1. B
2. B
3. C
4. A
5. C
6. A
7. C
8. D
9. B
10. B.

11. a) $(x_1, y_1) = (4, 6)$ $(x_2, y_2) = (1, -7)$ $m = -2$
 $n = 3$

$$P(x, y) = \left(\frac{nx_1 + mx_2}{m+n}, \frac{ny_1 + my_2}{m+n} \right)$$

$$= \frac{3(-4) - 2(1)}{1}, \frac{3(6) - 2(-7)}{1}$$

$$= (-14, 32)$$

2

b) $(3x+4)(x-5) - 2(x-5)^2 \geq 0$
 $(x-5)[3x+4 - 2(x-5)] \geq 0$
 $(x-5)[x+14] \geq 0$



$$x \leq -14 \text{ OR } x \geq 5 \quad x \neq 5$$

3

$$c) y = 2 \sin^{-1} \frac{x}{3}$$

$$D: -1 \leq \frac{x}{3} \leq 1$$

$$\boxed{-3 \leq x \leq 3}$$

$$R: -\frac{\pi}{2} \leq \frac{y}{2} \leq \frac{\pi}{2}$$

$$\boxed{-\pi \leq y \leq \pi}$$

$$d) \text{ let } y = (\sec x)^{\frac{1}{2}}$$

$$y = \frac{1}{(\cos x)^{\frac{1}{2}}}$$

$$y = (\cos x)^{-1/2}$$

$$\frac{dy}{dx} = -\frac{1}{2}(\cos x)^{-3/2} \cdot -\sin x \quad \text{OK.}$$

$$= \frac{\sin x}{2(\cos x)^{3/2}}$$

$$= \frac{\sin x}{2\sqrt{\cos^3 x}} = \left(\frac{1}{2} \tan x \sqrt{\sec x} \right)$$

$$e) \int_4^{16} \frac{1}{\sqrt{x}(4+x)} dx$$

$$u = \sqrt{x}$$

$$u = x^{\frac{1}{2}}$$

$$du = \frac{1}{2} x^{-1/2} dx$$

$$= 2 \int_4^{16} \frac{1}{2\sqrt{x}(4+x)} dx$$

$$du = \frac{1}{2\sqrt{x}} dx$$

$$= 2 \int_2^4 \frac{du}{4+u^2}$$

$$x=4, u=2$$

$$x=16, u=4$$

$$= 2 \times \frac{1}{2} \left[\tan^{-1} \frac{u}{2} \right]_2^4$$

$$= \tan^{-1} 2 - \tan^{-1} 1$$

$$= \tan^{-1} 2 - \pi/4$$

$$f) \int_0^{\pi/2} \cos^2 3x \, dx.$$

$$\int_0^{\pi/2} \frac{\cos 6x + 1}{2} \, dx$$

$$\frac{1}{2} \left[\frac{\sin 6x}{6} + x \right]_0^{\pi/2}$$

$$= \frac{1}{2} \left[\frac{\sin 3\pi}{6} + \frac{\pi}{2} - 0 \right]$$

$$= \frac{1}{2} \left[0 + \frac{\pi}{2} \right]$$

$$= \frac{\pi}{4}$$

$$\cos 6x = 2\cos^2 3x - 1$$

$$\frac{\cos 6x + 1}{2} = \cos^2 3x$$

⑫ a) Let $P(x) = x^2 - (a+2)x + 6$

$$P(a) = 0$$

$$0 = a^2 - (a+2)a + 6$$

$$0 = a^2 - a^2 - 2a + 6$$

$$2a = 6$$

$$a = 3$$

b) (i) $7\cos\theta - \sin\theta = R\cos(\theta + a)$
 $= R\cos\theta\cos a - R\sin\theta\sin a$

$$\therefore R\cos a = 7$$

$$R\sin a = 1$$

$$R^2 = 50$$

$$R = 5\sqrt{2}$$

$$\tan a = \frac{1}{7}$$

$$a = \tan^{-1} \frac{1}{7} = 8^\circ 8'$$

$$\therefore 7\cos\theta - \sin\theta = 5\sqrt{2}\cos(\theta + 8^\circ 8')$$

(ii) $5\sqrt{2}\cos(\theta + 8^\circ 8') = 5$

$$\cos(\theta + 8^\circ 8') = \frac{1}{\sqrt{2}}$$

$$\theta + 8^\circ 8' = 45^\circ \text{ or } 315^\circ$$

$$\theta = 36^\circ 52', 306^\circ 52'$$

c) Let $x = \sqrt[4]{m}$

$$\therefore x^4 = m$$

$$x^4 - m = 0$$

$$\text{Let } f(x) = x^4 - m$$

$$f'(x) = 4x^3$$

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$x_2 = x_1 - \frac{x_1^4 - m}{4x_1^3}$$

$$x_2 = \frac{4x_1^4 - x_1^4 + m}{4x_1^3}$$

$$x_2 = \frac{3x_1^4 + m}{4x_1^3}$$

$$d) \frac{d^2x}{dt^2} = 2x - 3$$

$$\frac{d}{dx}\left(\frac{1}{2}v^2\right) = \ddot{x}$$

$$\frac{1}{2}v^2 = \int (2x - 3) dx$$

$$\frac{1}{2}v^2 = x^2 - 3x + C$$

$$v = 0, x = 4$$

$$0 = 16 - 12 + C$$

$$\therefore C = -4$$

$$\frac{1}{2}v^2 = x^2 - 3x - 4$$

$$v^2 = 2(x^2 - 3x - 4)$$

$$(i) \text{ Let } x = 0$$

$$v^2 = 2(0 - 0 - 4)$$

$$v^2 = -8$$

which is not poss.

$\therefore$ Particle does not pass through origin.

$$(iii) 10^2 = 2(x^2 - 3x - 4)$$

$$50 = x^2 - 3x - 4$$

$$0 = x^2 - 3x - 54$$

$$0 = (x - 9)(x + 6)$$

$$\therefore x = 9 \text{ or } x = -6.$$

Since particle starts at $x = 4$ from rest and does not pass through origin, then only solution is $x = 9$.

(iv) Step 1. Prove true for $n = 1$

$$1(1 + 3) = 4 \text{ which is div. by } 2$$

$\therefore$ True for $n = 1$

Step 2: Assume true for $n = k$

$$\text{ie. } k(k + 3) = 2m \text{ for a pos integer } m$$

$$\text{ie. } k^2 + 3k = 2m$$

Step 3 : Prove true for $n = k+1$

ie. $(k+1)(k+4)$ is div by 2

$$(k+1)(k+4) = k^2 + 5k + 4$$

$$= k^2 + 3k + 2k + 4$$

$$= 2m + 2k + 4 \quad \text{from Step 2 assumption}$$

$$= 2(m+k+2)$$

$\therefore$ True for $n = k+1$

Step 4 : By principle of mathematical induction
true for all positive integers n .

13. a)

$$\int \frac{dx}{\sqrt{9-4x^2}}$$

$$\int \frac{dx}{\sqrt{4(\frac{9}{4}-x^2)}}$$

$$= \frac{1}{2} \int \frac{dx}{\sqrt{\frac{9}{4}-x^2}}$$

$$= \frac{1}{2} \sin^{-1} \frac{2x}{3} + C$$

b) $\frac{dN}{dt} \propto N$

$$\frac{dN}{dt} = k(N-400)$$

i) $N = 400 + Ae^{kt}$

$$\frac{dN}{dt} = k \cdot Ae^{kt}$$

$$= k(N-400) \quad \text{since } N = 400 + Ae^{kt}$$

$$Ae^{kt} = N - 400$$

(ii) $t=0, N=1000$

$t=12, N=900$

$t=?, N=500$

$$N = 400 + Ae^{kt}$$

$$1000 = 400 + Ae^0$$

$$A = 600$$

$$\therefore N = 400 + 600e^{kt}$$

$$900 = 400 + 600e^{12k}$$

$$\frac{500}{600} = e^{12k}$$

$$600$$

$$\frac{1}{12} \ln \frac{5}{6} = k$$

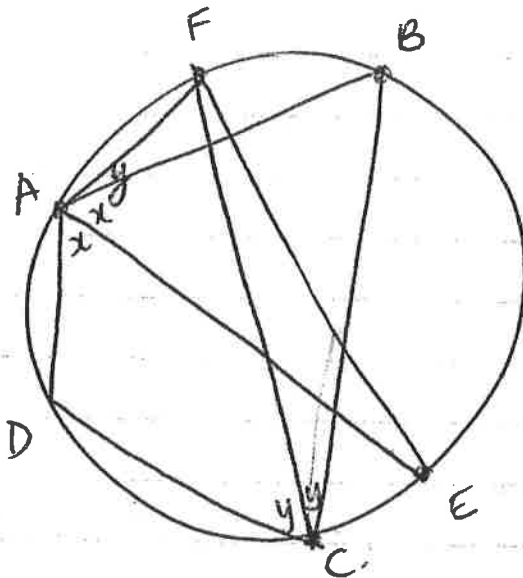
$$\begin{aligned} \therefore N &= 400 + 600 e^{\frac{1}{12} \ln \frac{5}{6} t} \\ 500 &= 400 + 600 e^{\frac{1}{12} \ln \frac{5}{6} t} \\ \frac{100}{600} &= e^{\frac{1}{12} \ln \frac{5}{6} t} \end{aligned}$$

$$\ln \frac{1}{6} = \frac{t}{12} \ln \frac{5}{6}$$

$$\frac{12 \ln \frac{1}{6}}{\ln \frac{5}{6}} = t \quad t = 117.9 \text{ weeks}$$

2

9



Let $\angle DAE = \angle BAE = x$

and $\angle FCD = \angle FCB = y$.

$2x + 2y = 180$ (opposite angles of cyclic quadrilateral are supplementary)
Join AF $\therefore x + y = 90$.

$\angle BCF = \angle FAB = y$ (angles in same segment standing on same arc are equal)

$$\therefore \angle FAE = 2x + y$$

$= 90^\circ \therefore FE$ is diameter (angle in a semicircle is 90°).

d) ii) If right angled then from converse of Pyth. Thm.

$$XY^2 = XT^2 + TY^2$$

$$LHS = 25^2$$

$$= 625$$

$$RHS = 15^2 + 20^2$$

$$= 225 + 400$$

$$= 625$$

$$LHS = RHS$$

1

$$(ii) \tan \theta = \frac{20}{15} = \frac{4}{3}$$

$$\theta = \tan^{-1} \frac{4}{3}$$

$$\therefore \sin \theta = \frac{h}{15}$$

$$h = 15 \sin(\tan^{-1} \frac{4}{3})$$

$$h = 12$$

OK using similar Δ 's.

2

(iii) Let the angle be θ

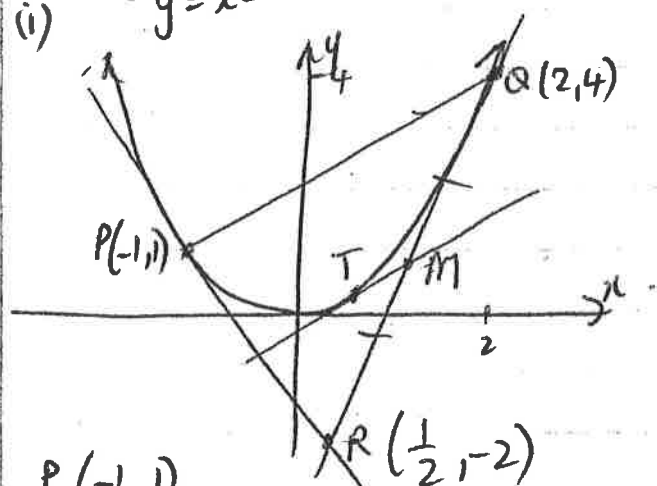
$$\tan \theta = \frac{12}{5}$$

$$\theta = \tan^{-1} \frac{12}{5}$$

$$= 67^\circ 23'$$

Q) $x=t, y=t^2$

(i) $y=x^2$



$P(-1, 1)$

$Q(2, 4)$

(ii) $y=x^2$

$\frac{dy}{dx} = 2x$

At $x = -1$

$\frac{dy}{dx} = -2$

At $x = 2$

$\frac{dy}{dx} = 4$

mo of PR : $y-1 = -2(x+1)$

$y = -2x - 1$

mo of QR : $y-4 = 4(x-2)$

$y-4 = 4x-8$

$y = 4x - 4$

$y = -2x - 1$

$y = 4x - 4$

$$\therefore 4x - 4 = -2x - 1$$

$$6x = 3$$

$$x = \frac{1}{2}$$

$$\therefore y = -2$$

$$\therefore R\left(\frac{1}{2}, -2\right)$$

2.

$$(iii) \frac{dy}{dx} = 2x$$

$$\text{At } T(t, t^2)$$

$$\frac{dy}{dx} = 2t$$

Eqn of tang. at T.

$$y = 2tx - t^2$$

Let M = Midpt of R

$$= \left(\frac{2 + \frac{1}{2}}{2}, \frac{4 - 2}{2} \right)$$

$$= \left(\frac{5}{4}, 1 \right)$$

Sub M into eqn of tang. at T.

$$1 = 2t\left(\frac{5}{4}\right) - t^2$$

$$1 = \frac{5t}{2} - t^2$$

$$2 = 5t - 2t^2$$

$$2t^2 - 5t + 2 = 0$$

$$2t \times 1$$

$$t \times 2$$

$$(2t - 1)(t - 2) = 0$$

$$\therefore t = \frac{1}{2}, t = 2$$

But Q, $t = 2$

$$\therefore T: t = \frac{1}{2}$$

But m of tangent

$$= 2t$$

$$= 1$$

$$m \text{ of } PQ = 4 - 1/2 + 1 = 1$$

$\therefore$ tangent at T is
parallel to PQ .

2

(14)

a) (i) $x = \frac{3(\cos 2t - \sin 2t)}{\sqrt{2}}$

$$\dot{x} = \frac{3}{\sqrt{2}} (-2\sin 2t - 2\cos 2t)$$

$$\ddot{x} = \frac{3}{\sqrt{2}} (-4\cos 2t + 4\sin 2t)$$

$$= -4 \cdot \frac{3}{\sqrt{2}} (\cos 2t - \sin 2t)$$

$\ddot{x} = -4x$ which takes the form $\ddot{x} = -n^2 x$
where $n=2$

2

(ii) $T = \frac{2\pi}{n} = \frac{2\pi}{2} = \pi$ seconds

1

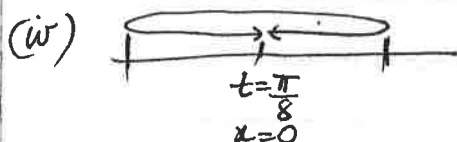
(iii) Sub $t = \frac{\pi}{8}$ into x

$$x = \frac{3(\cos \frac{\pi}{4} - \sin \frac{\pi}{4})}{\sqrt{2}}$$

$$x = \frac{3}{\sqrt{2}} \left(\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

1

$$x = 0$$



$\frac{\pi}{4}$ seconds to
go from centre
to endpt. as
period $= \pi$.

Amplitude occurs at

$$t = \frac{\pi}{8} + \frac{\pi}{4} = \frac{3\pi}{8}$$

(since $x=0$ is centre of motion)

Sub $t = \frac{3\pi}{8}$ into x

$$x = \frac{3}{\sqrt{2}} \left(\cos \frac{3\pi}{4} - \sin \frac{3\pi}{4} \right)$$

$$= \frac{3}{\sqrt{2}} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

$$= \frac{3}{\sqrt{2}} \times -\frac{2}{\sqrt{2}} = -3 \therefore \text{Amplitude} = 3$$

2

OK let $v=0$

$$\frac{3}{\sqrt{2}}(-2\sin 2t - 2\cos 2t) = 0$$

$$-2\sin 2t = 2\cos 2t$$

$$\tan 2t = -1$$

$$2t = \pi - \frac{\pi}{4}, 2\pi - \frac{\pi}{4}$$

$$2t = \frac{3\pi}{4}, \frac{7\pi}{4}$$

$$t = \frac{3\pi}{8}, \frac{7\pi}{8}$$

$$\text{Sub } t = \frac{3\pi}{8}$$

$$x = \frac{3}{\sqrt{2}} \left(\cos \frac{3\pi}{4} - \sin \frac{3\pi}{4} \right)$$

$$x = \frac{3}{\sqrt{2}} \left(-\frac{1}{\sqrt{2}} - \frac{1}{\sqrt{2}} \right)$$

$$x = \frac{3}{\sqrt{2}} \cdot -\frac{2}{\sqrt{2}}$$

$$x = -2 \quad \text{Amplitude} = 3$$

(b) (i) In $\triangle PBC$

, Let $\angle BPC = \alpha$

$$\tan \alpha = \frac{20}{x}$$

tan In $\triangle APB$

$$\tan (\alpha + \theta) = \frac{32}{x}$$

$$\frac{\tan \alpha + \tan \theta}{1 - \tan \alpha \tan \theta} = \frac{32}{x}$$

$$\frac{\frac{20}{x} + \tan \theta}{1 - \frac{20}{x} \cdot \tan \theta} = \frac{32}{x}$$

$$\frac{\frac{20 + x \tan \theta}{x}}{\frac{x - 20 \tan \theta}{x}} = \frac{32}{x}$$

$$x(20 + x \tan \theta) = 32(x - 20 \tan \theta)$$

$$20x + x^2 \tan \theta = 32x - 640 \tan \theta$$

$$x^2 \tan \theta - 12x + 640 \tan \theta = 0.$$

$$\tan \theta (x^2 + 640) = 12x$$

$$\tan \theta = \frac{12x}{x^2 + 640}$$

$$\theta = \tan^{-1} \left(\frac{12x}{x^2 + 640} \right)$$

$$\begin{aligned} \text{(ii)} \quad \frac{d\theta}{dx} &= \frac{1}{1 + \frac{(12x)^2}{(x^2 + 640)^2}} \times \frac{(x^2 + 640) \cdot 12 - 12x(2x)}{(x^2 + 640)^2} \\ &= \frac{(x^2 + 640)^2}{(x^2 + 640)^2 + 144x^2} \times \frac{12(x^2 + 640 - 2x^2)}{(x^2 + 640)^2} \\ &= \frac{12(640 - x^2)}{(x^2 + 640)^2 + 144x^2} \end{aligned}$$

$$\text{let } \frac{d\theta}{dx} = 0$$

$$12(640 - x^2) = 0.$$

$$x^2 = 640$$

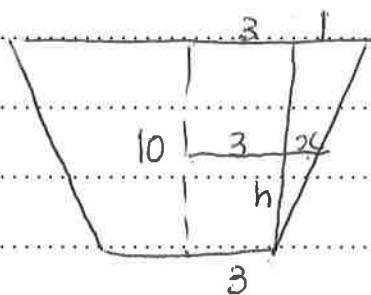
$$x = 8\sqrt{10} \quad x \geq 0$$

$$\text{Test } x = 8\sqrt{10}$$

x	25	$8\sqrt{10}$	26
$\frac{d\theta}{dx}$	12×15	0	12×-36
	+		-

Sign change $\searrow \therefore \text{max at } x = 8\sqrt{10}$

(i)



Using similar triangles

$$\frac{x}{1} = \frac{h}{10}$$

$$x = \frac{h}{10}$$

$$H = h$$

$$a = 3 + \frac{h}{10}$$

$$b = 3$$

$$V = \frac{\sqrt{3}}{2} \left(\left(3 + \frac{h}{10} \right)^2 + 3 \left(3 + \frac{h}{10} \right) + 9 \right) \cdot h$$

$$= \frac{h\sqrt{3}}{2} \left(9 + \frac{6h}{10} + \frac{h^2}{100} + 9 + \frac{3h}{10} + 9 \right)$$

$$= \frac{h\sqrt{3}}{2} \left(\frac{900 + 60h + h^2 + 900 + 30h + 900}{100} \right)$$

$$= \frac{h\sqrt{3}}{2} \left(\frac{2700 + 90h + h^2}{100} \right)$$

$$= \frac{\sqrt{3}}{2} \left(\frac{2700h + 90h^2 + h^3}{100} \right) \times \frac{10}{10}$$

$$V = \frac{\sqrt{3}}{2000} (27000h + 900h^2 + 10h^3)$$

$$(ii) \frac{dV}{dt} = 7$$

$$\frac{dh}{dt} = \frac{dV}{dt} \times \frac{dh}{dV}$$

$$\frac{dV}{dh} = \frac{\sqrt{3}}{2000} (27000 + 1800h + 30h^2)$$

$$\frac{dh}{dt} = 7 \times \frac{2000}{\sqrt{3} (27000 + 1800h + 30h^2)}$$

When $h = 5$:

$$\frac{dh}{dt} = 7 \times \frac{2000}{\sqrt{3} (27000 + 1800 \times 5 + 30 \times 25)}$$

$$= \frac{14000}{\sqrt{3} (36750)}$$

$$= \frac{8}{\sqrt{3} \times 21} \times \frac{\sqrt{3}}{\sqrt{3}}$$

$$= \frac{8\sqrt{3}}{63} \text{ cm/s}$$

$$= 0.2199 \text{ cm/s}$$